«part def»

<CDT-PC-01> Custom Candidate Operating Procedures

«part»
tailorAndAcceptModel : 'CSRM Tailoring and Model Acceptance Procedure'

«part»
verifyBatteryAwareScenarios : 'Battery-Aware Pre-Physical Verification Procedure'

«part»
releaseAndReviewBaseline : 'Baseline Release and Collaborative Review Procedure'

«part»
synchronizeSysMLAndOpenUSD : 'SysML and OpenUSD Synchronization Procedure'

«part»
executeAndReproduceAnalysis : 'Controlled Analysis Execution and Reproduction Procedure'

«part def»

<CDT-PN-01> CSRM and SysML v2 Model Gateway Physical Node

doc

Candidate-owned gateway for CSRM semantics, textual SysML v2 exchanges, stable identity, and participant review. The authoritative mission model remains the external Versioned System Model defined by the black-box context.

«part»
host : 'Model Gateway Compute Host'

«part»
csrmLibrary : 'CSRM Semantic Reference Library'

«part»
baselineClient : 'Version Control and Baseline Client'

«part»
sysmlAdapter : 'SysML v2 Model API and Text Adapter'

«part»
identityRegistry : 'Stable Identity Trace and Mapping Registry'

«part»
commandAndReview : 'Framework Command and Review Interface'

«part»
modelDevelopmentAdapter : 'Model Development Interoperability Adapter'

«part»
requirementsAdapter : 'Requirements Management Interoperability Adapter'

«part def»

<CDT-PN-02> Python Digital Thread and Analysis Runner Physical Node

doc

Candidate-owned orchestration spine. Detailed engineering calculations remain in external SA tools, while this node controls their exchange, execution identity, impact analysis, acceptance checks, and evidence packaging.

«part»
host : 'Python Runner Compute Host'

«part»
workflow : 'Controlled Workflow Orchestrator'

«part»
evaluationRunner : 'Evaluation Manifest Runner'

«part»
transferChecker : 'Transfer and Semantic Checker'

«part»
runtime : 'Locked Python Runtime and Environment'

«part»
analysisBridge : 'Analysis Request and Result Bridge'

«part»
exchangePackage : 'Typed Analysis Exchange Package'

«part»
evidencePackager : 'Evidence and Provenance Packager'

«part»
impactAnalyzer : 'Dependency and Change Impact Analyzer'

«part»
adapterRegistry : 'Simulation and Analysis Adapter Registry'

«part def»

<CDT-PN-03> Omniverse and Isaac Interoperability Physical Node

doc

Candidate extensions and controlled data deployed with the external Omniverse and Isaac environment. Omniverse supplies spatial collaboration and visualization. Isaac supplies a robotics scenario environment. Specialized EPS, orbit, circuit, thermal, and battery solvers remain external SA capabilities connected through Python or FMI adapters.

«part»
host : 'Omniverse Integration Compute Host'

«part»
fmiBridge : 'FMI and FMU Co-Simulation Bridge'

«part»
twinPackage : 'Controlled OpenUSD Twin Package'

«part»
evidenceCollector : 'Simulation Evidence Collector'

«part»
dataBinder : 'Telemetry and Simulation Data Binder'

«part»
isaacAdapter : 'Isaac Sim Controlled Scenario Adapter'

«part»
epsBatteryConnector : 'EPS and Battery Model Connector'

«part»
semanticBridge : 'SysML v2 to OpenUSD Semantic Bridge'

«part»
scenarioLibrary : 'Battery-Aware Verification Scenario Library'

«part»
collaborationExtension : 'Omniverse Kit Collaboration Extension'